Utility of low-cost sensor measurement for predicting ambient PM2.5 concentrations: Evidence from a monitoring network in Accra, Ghana

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Introduction

Ambient air pollution, particularly particulate matter (PM2.5), is linked to serious health issues such as respiratory infections, COPD, cardiovascular diseases, lung cancer, and adverse birth outcomes. The WHO estimates 7 million annual deaths due to air pollution. In sub-Saharan Africa (SSA), air quality has deteriorated due to urbanization, increased vehicle use, and industrialization, with PM2.5 levels in SSA cities often exceeding 100 µg/m³.

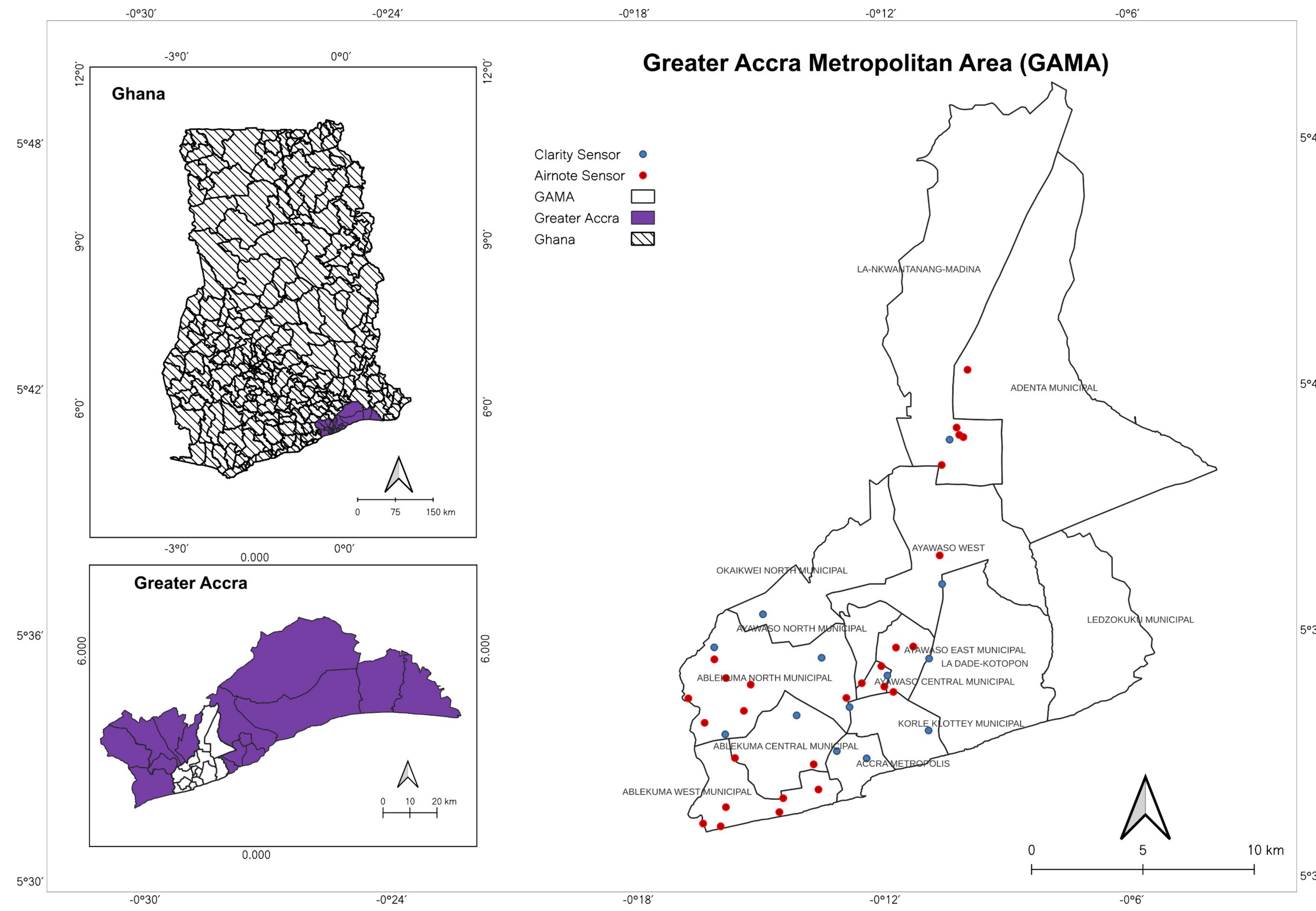
Challenges in Monitoring:

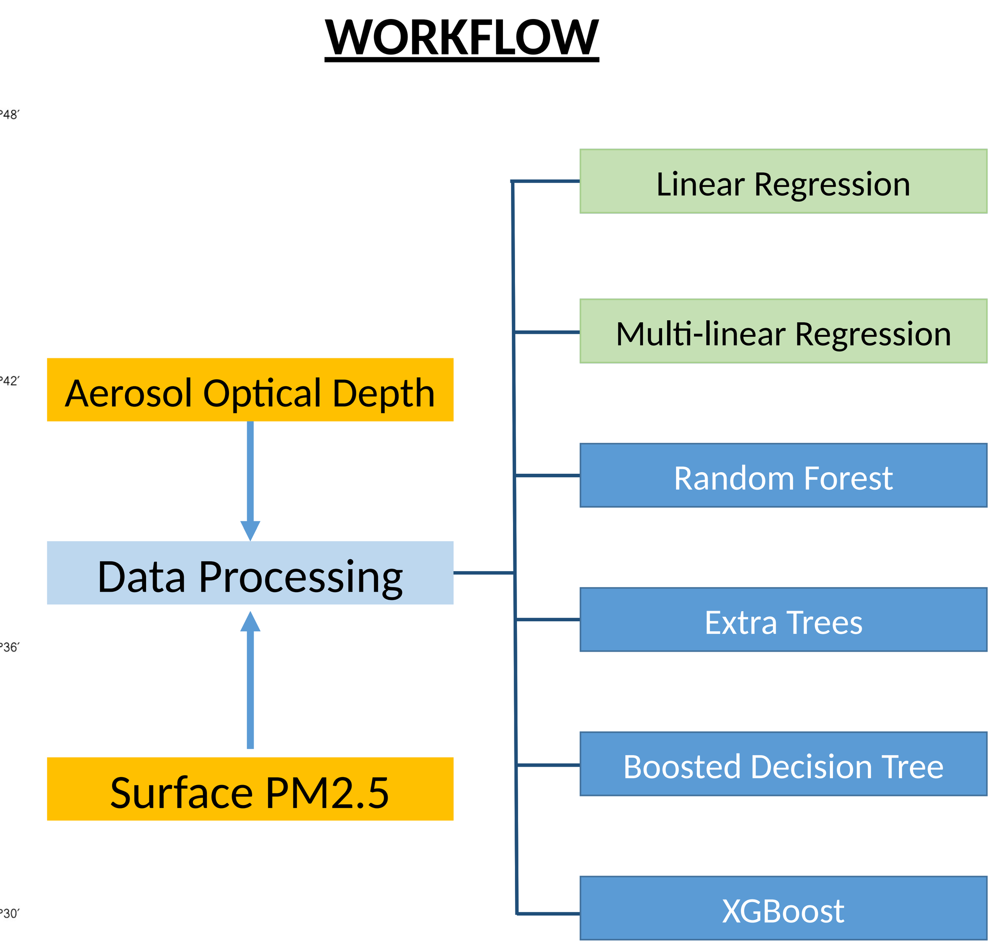
Air quality monitoring in SSA is limited. Satellite aerosol remote sensing helps estimate ground-level PM2.5 but has significant limitations, including diurnal variability, cloud cover, and uncertainties in aerosol optical depth (AOD) correlations.

Emerging Solutions:

Low-cost air quality sensors offer high spatiotemporal resolution, potentially filling data gaps in SSA. These sensors can complement satellite data, providing more reliable and real-time air quality information. We evaluate the efficacy of low-cost sensors in Accra, Ghana, by comparing PM2.5 measurements with satellite-derived AOD data, aiming to enhance air pollution monitoring in regions with limited resources.

Methodology





Results

Table 1. Comparison of PM2.5 Measurements from Clarity and Airnote Sensors with MODIS Satellite Data during overpass time.

	Total PM2.5 Measurement	Paired PM2.5 Measurement with MODIS Optical Depth	Percentage Difference
Clarity Sensor	11,278	1642	14.6%
Airnote Sensor	9,979	869	8.7%

Table 2. Ordinary Least Squares (OLS) regression prediction(PM2.5 measurements – Optical Depth 550nm)

	Clarity (Raw PM2.5)	Clarity(Corrected PM2.5)	Airnote Sensor
β Coefficient (95% CI)	19.41(17.21,22.26)	8.58 (7.56, 9.59)	20.04 (17.43, 22.65)
R- Squared	0.196	0.182	0.252
Adjusted R Squared	0.196	0.181	0.251
RMSE	16.59	7.50	13.72
Moran's I test	0.099	0.017	0.00020
p(Moran's I test)	1	1	0.756

Chart 1. R^2 Model comparison results – Clarity S Node (Uncorrected PM2.5 Measurement)

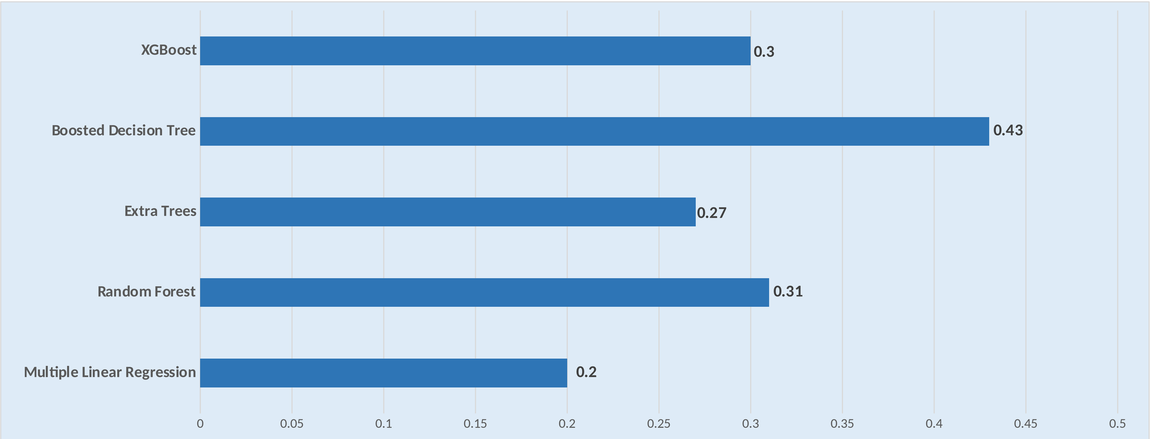


Chart 2. R^2 Model comparison results – Clarity S Node (Corrected PM2.5 Measurement)

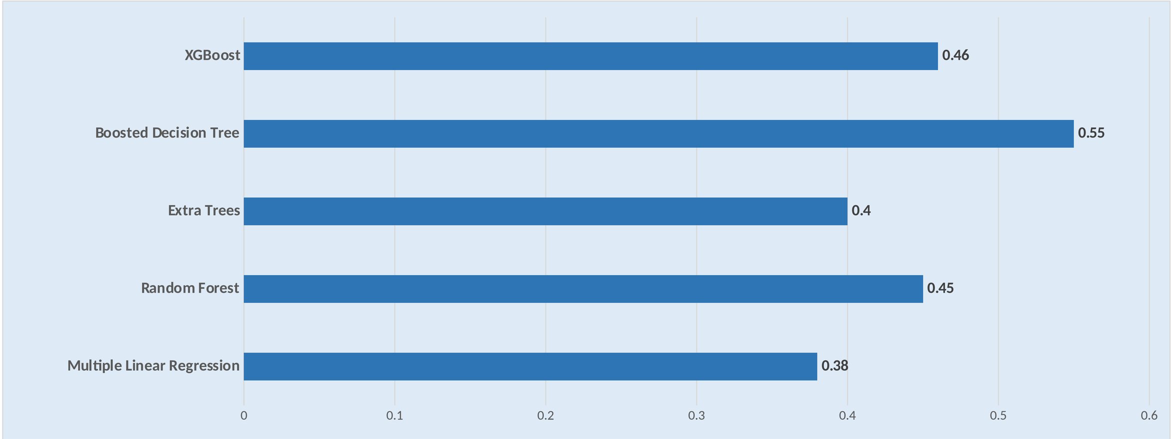
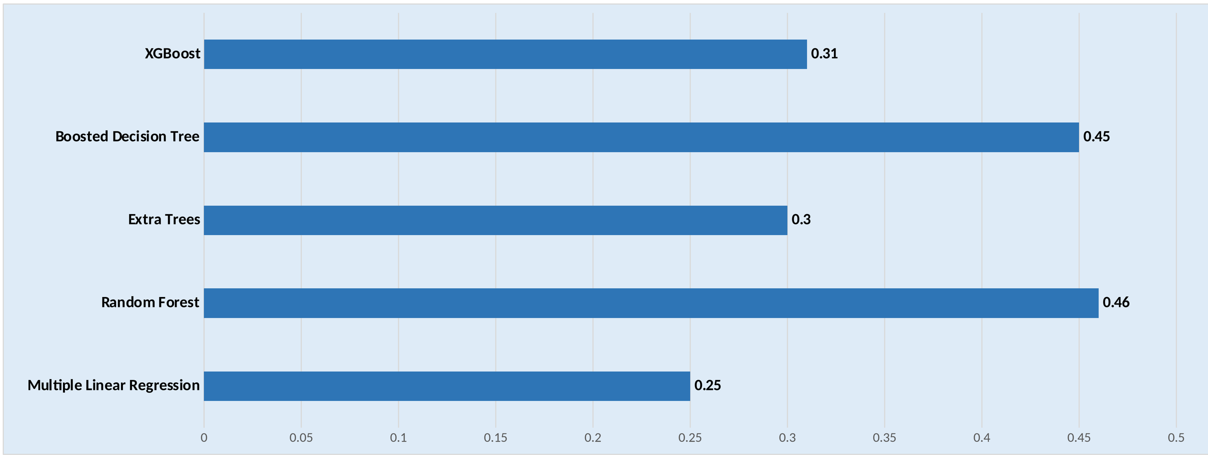


Chart 3. R^2 Model comparison results (Airnote Sensor)



Discussion

The simple linear regression showed statistically significant positive relationships, with AOD increases corresponding to higher PM2.5 readings. However, the low R^2 values (0.19-0.25) suggest that AOD explains only a small portion of the PM2.5 variation, indicating that OLS regression is not the most effective model for this relationship. Residual analysis and Moran's I test further revealed model bias and non-linearity.

Meteorological factors such as temperature and humidity were incorporated into the models, slightly improving R^2 values. Machine learning models (Random Forest, XGBoost) also showed better performance, though still within the low-to-moderate range. These models can capture more complex relationships but require careful feature engineering.

The use of low-cost sensors poses challenges, particularly in calibration against reference-grade monitors. While correction factors were applied, the reliability of adjusted data remains uncertain, with potential systematic errors due to spatial and temporal variations. The performance of optical particle sensors is influenced by particle properties and atmospheric conditions, complicating accurate PM2.5 estimation.

Conclusions

It is evident in this research that despite the use of corrected measurement, the predictability of the ambient PM2.5 from satellite derived AOD and meteorological data from low-cost sensors was relatively poor. This assessment indicates a few things; that ground level PM2.5 is influence by other factors and hence complex modelling techniques needs to be employed for an accurate predictive model and also the algorithms for the calibration of these low-cost sensors needs to be evaluated during each study to improve on accuracy of the correction factor.

Acknowledgment

The study was funded by Clean Air Fund and NIH under the DSI-Africa program. We are also grateful to all the 13 beneficiary assemblies of the Breathe Accra project through whose support we were able to set up the monitoring network.